

## No. of Functional Features Included in the Solution

|               |                                                              |
|---------------|--------------------------------------------------------------|
| Date          | 04-07-2026                                                   |
| Team ID       |                                                              |
| Project Name  | Smart Agricultural Production Optimization Engine (OptiCrop) |
| Maximum Marks | 5 Marks                                                      |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | Feature Name                | Feature Description                                                              | Module / Component                      | Status | Marks Contribution |
|------|-----------------------------|----------------------------------------------------------------------------------|-----------------------------------------|--------|--------------------|
| 1    | User Input Form             | Accepts Nitrogen, Phosphorus, Potassium, temperature, humidity, pH, and rainfall | Flask HTML Form (findyourcrop.html)     | Done   | High               |
| 2    | Data Validation             | Validates and type-casts user input before prediction                            | Flask Backend (app.py)                  | Done   | Medium             |
| 3    | Crop Prediction             | Predicts the most suitable crop using the trained ML model                       | Prediction Engine (Logistic Regression) | Done   | High               |
| 4    | ML Model Loading            | Loads the trained model and scaler (.pkl files) at startup                       | Model Loader                            | Done   | High               |
| 5    | Crop Recommendation Display | Displays the predicted crop to the user                                          | Output Module (result.html)             | Done   | Medium             |
| 6    | Dataset Integration         | Uses the agricultural dataset for training and prediction                        | Dataset Loader                          | Done   | Medium             |
| 7    | GitHub Repository           | Source code managed using GitHub                                                 | Version Control                         | Done   | Low                |
| 8    | Flask Deployment            | Runs the application using Flask, deployed on Render                             | Application Interface                   | Done   | Medium             |

### Feature Summary:

| Metric                             | Count / Value                                                                                    |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| Total Features Planned             | 8                                                                                                |
| Total Features Implemented         | 8                                                                                                |
| Core / Must-Have Features          | Crop Prediction, User Input Form, ML Model Loading, Data Validation, Crop Recommendation Display |
| Additional / Nice-to-Have Features | GitHub Repository, Flask Multi-Page Interface (Home / About), Dataset Integration                |
| Features Tested & Verified         | 8 / 8 (100%)                                                                                     |

### Feature Category Breakdown:

| S.No | Category                  | Features in Category | Example Features                                                |
|------|---------------------------|----------------------|-----------------------------------------------------------------|
| 1    | User Interface (UI)       | 3                    | Home page, Find Your Crop input form, Prediction result display |
| 2    | Backend / Logic           | 3                    | Crop prediction model, Input validation, Model loading          |
| 3    | Database / Storage        | 1                    | Crop dataset (CSV), Trained model (.pkl)                        |
| 4    | API / Integration         | 1                    | Flask routes integrating with the ML model                      |
| 5    | Security / Authentication | 0                    | Not implemented                                                 |